VidyaLok

- Where Curiosity Meets Clarity

Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
(Approved by AICTE New Delhi & Govt. of Maharashtra, Affiliated to University of Mumbai)
(Religious Jain Minority)

Department of Computer Science & Engineering (AI & ML)

VidyaLok: An AI-Powered Smart Library Ecosystem using NLP and Predictive Models for APSIT

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Under the Guidance
~ Prof. Vijesh Nair

Introduction

- In an era where knowledge is the true currency of innovation, VidyaLok emerges as more than just a library system — it is a digital gateway to the academic soul of APSIT. Purpose-built for our newly established library, VidyaLok combines the power of Artificial Intelligence, real-time analytics, and seamless web technologies to deliver a deeply personalized, engaging, and intelligent library experience. From auto-tracking student entries and space occupancy to recommending the right resources at the right time, VidyaLok is designed to serve not just as a management tool, but as a living, breathing knowledge ecosystem. With students, faculty, and administrators in mind, this project isn't just an academic requirement — it's a real-world solution that the entire campus can rely on, grow with, and be proud of.

Problem Statement

- Quickly find relevant books due to the absence of **intelligent search and personalized recommendations**.
- Get **round-the-clock support**, as no chatbot exists for query resolution.
- **Tracking entry/exit and occupancy** accurately.
- **Analyzing student feedback** to improve services.
- **Broadcasting urgent notifications** during emergencies.

Objective

The objective of **VidyaLok** is to develop a comprehensive, web-based Library Management and Engagement System for APSIT' s newly established library that simplifies and digitizes daily operations for both students and faculty. The system will:

- ***Digitize core library functions*** like book cataloging, issue/return tracking, and student records
- ***Auto-log student entry/exit*** using their unique Student ID and maintain real-time occupancy data
- ***Track seat availability dynamically*** to help manage library space effectively
- ***Provide students with personalized dashboards*** for tracking borrowed books, due dates, and suggested readings
- ***Enable faculty/admins to upload notes, manage inventory***, and monitor usage insights
- ***Integrate ML for smart recommendations*** based on user interests, course, and past activity
- Offer a ***real-time, scalable, and user-friendly system*** for the entire college

Literature Survey

Year	Title	Domain	Algorithm	Dataset	Gap
2025	AI-Powered Library Management System	Library Systems (Automation/ NLP)	NLP search, content-based and recommendations, IoT/RFID integration	N/A (theoretical framework)	Traditional library processes are manual and slow; need AI to automate tasks, personalize recommendations, and improve efficiency
2025	Sentiment Analysis in Learning Management Systems: Understanding Student Feedback at Scale	EdTech (NLP/ Sentiment	Word embeddings + LSTM with attention; logistic regression baseline	RateMy-Professor student feedback	Sentiment analysis of student feedback in LMS is under-explored; existing systems lack tools to interpret qualitative feedback and bridge communication gaps
2024	Lib-Bot: A Smart Librarian Chatbot Assistan	Library Services (Chatbot/NLP)	BERT for intent classification; speech-to-text; Naïve Bayes/SVM fallback	Library FAQ and query logs	Users often cannot reach librarians); need an AI chatbot to handle common inquiries via text/voice.

Literature Survey

Year	Title	Domain	Algorithm	Dataset	Gap
2024	QR Code Integrated ML Analytics Library System	Smart Library (IoT/Smart Campus)	Unique QR-code mapping + ML rule-based system	N/A (conceptual)	Need of touchless library solution. Proposed system assigns unique QR IDs to users, enabling touch-free check in/out and simplified inventory with minimal overhead
2022	A Hybrid Book Recommendation System for University Library	Library/ Recommender	Hybrid: content based + collaborative filtering	University library catalog + course syllabus data	Growing library collections make manual search slow and unsatisfying; existing systems lack recommendation features. This hybrid model leverages course syllabi to address cold-start and improve book discovery for users
2021	Library Book Intelligent Recommendation System Based on Artificial Intelligence	Library/ Recommender	Hybrid CF (user based + item based clustering)	User book ratings/ evaluation	Traditional recommenders yield higher errors in sparse data. This AI-based system cleans and clusters users, applying hybrid collaborative

Limitation of existing system

- **Lack of Personalization –** All students are treated the same, without considering their branch, semester, or interests.
- **Basic Search Mechanism –** Systems rely only on keyword matching, not on semantic or intelligent search.
- **Manual Entry/Exit Tracking –** Attendance is logged manually or via RFID, which is error-prone and gives no real-time analytics.
- **No Interactive Assistance –** No chatbot or virtual help for students outside library hours.
- **No Automated Summarization –** Students must read lengthy chapters/research papers in full.
- **Feedback Not Analyzed –** Suggestions/complaints are collected but not processed intelligently.

Uniqueness

- AI Chatbot for Natural Queries
- Emergency Broadcasting & Smart Notifications
- Auto-Summarized Insights
- QR-Based Auto Entry/Exit Logging
- Live Seat Availability System
- Department-Wise Usage & Demand Graphs
- Automated Weekly Reports (PDF)

System Overview

👨 🎓 For Students:

- 🪪 **Student ID-Based Entry System** – Scan ID to log entry/exit and access personal data
- 📚 **Smart Book Search & Borrow System** – Search books, check availability, request issue
- **Issue & Return History** Request New Book, Library Chatbot, Entry Exit Log Viewer
- 🧾 **Borrowing History & Due Reminders** – Personalized dashboard with due dates, return alerts
- 🧠 **AI-Based Book Recommendations** – Suggests relevant books based on branch, semester, interest
- 🪑 **Live Seat Availability Indicator** – Know how many seats are vacant in the library in real-time
- **Library Leaderboard / Badging System** *(Optional, fun feature)*

👩 💼 For Admins & Faculty:

- 🧾 **Book Inventory Management** – Add/edit/remove books, manage stock, track damaged/lost items
- ⏱️ **Real-Time Entry/Exit Logs** – View which students entered, when, and how long they stayed
- 📊 **Admin / Analytics Dashboard** – Monitor book demand, peak usage hours, active students
- 📤 **Borrow & Return Management , Live Occupancy Monitor**
- 🔔 **Overdue Alerts & Fine Reports** – Automated reports for late returns
- **Emails Alert to Borrower of Books**
- Weekly/Monthly Usage Report Generator (PDF)
- Emergency Broadcast / Alert System
- Department Wise Book Usage Graph

🤖 AI / ML-Powered Smart Features:

- 🧠 **Book Recommendation Engine** – ML model trained on student behavior, course curriculum, and interest tags (Collaborative Filtering & TF-IDF & Siamese-BERT, sentence embeddings, or Fine-tuned Transformers).
- 🧍 ♂️ **Occupancy Heatmap & Space Insights** – Visual analytics of busy hours
- 🗣️ **Intelligent AI Chatbot -** An **AI-powered chatbot** that understands **natural language queries** from students and recommends the most relevant book available in the library (Top K Book Fetched).
- 📈**Popularity Predictive Analysis** – See which books will likely be in high demand during exam time
- AI-Powered Book Summarization Engine**:** Chapters Wise Summary of Each Book.
- **Library Sentiment Tracker -** Analyzes feedback/comments using sentiment analysis to measure student satisfaction over time.
- **AI-Generated Practice Questions** – Automatically creates 3–5 MCQs from each book.

Tech Stack / Tools

- **Frontend**

Html, CSS, JavaScript, Typescript.js, Tailwind CSS

- **Backend**

Node.js, Express.js

- **Databases**

MongoDB, PostgreSQL

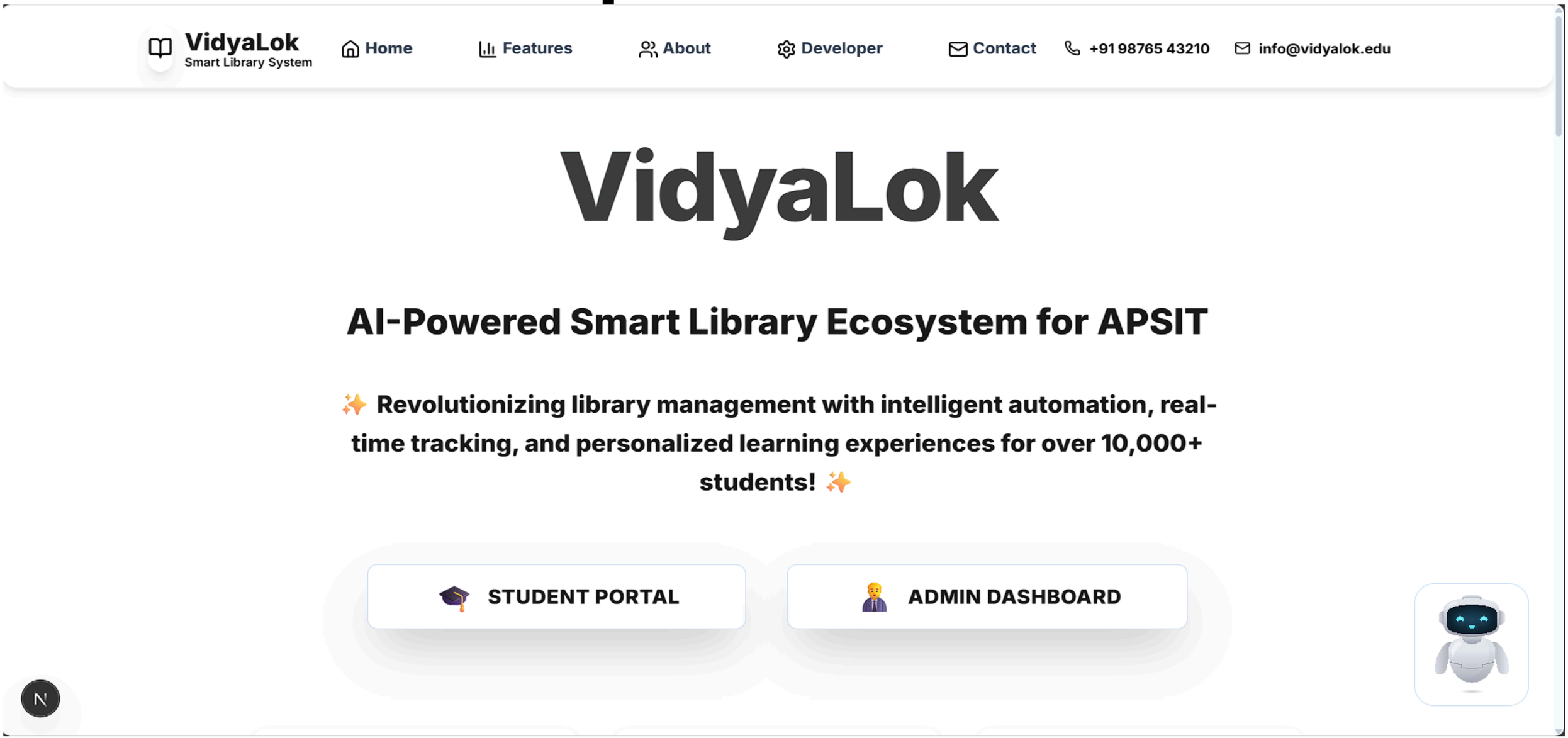
- **Machine Learning**

Python, Scikit-learn, TensorFlow, Vector DB, Embedding (Sentence-BERT / OpenAI Embed)

Search & Recommendation TF-IDF, cosine similarity, NLP

- **Notifications –** SMTP (GMAIL), Twilio
- **Authentication –** JWT + Role-Based Auth

Implementation



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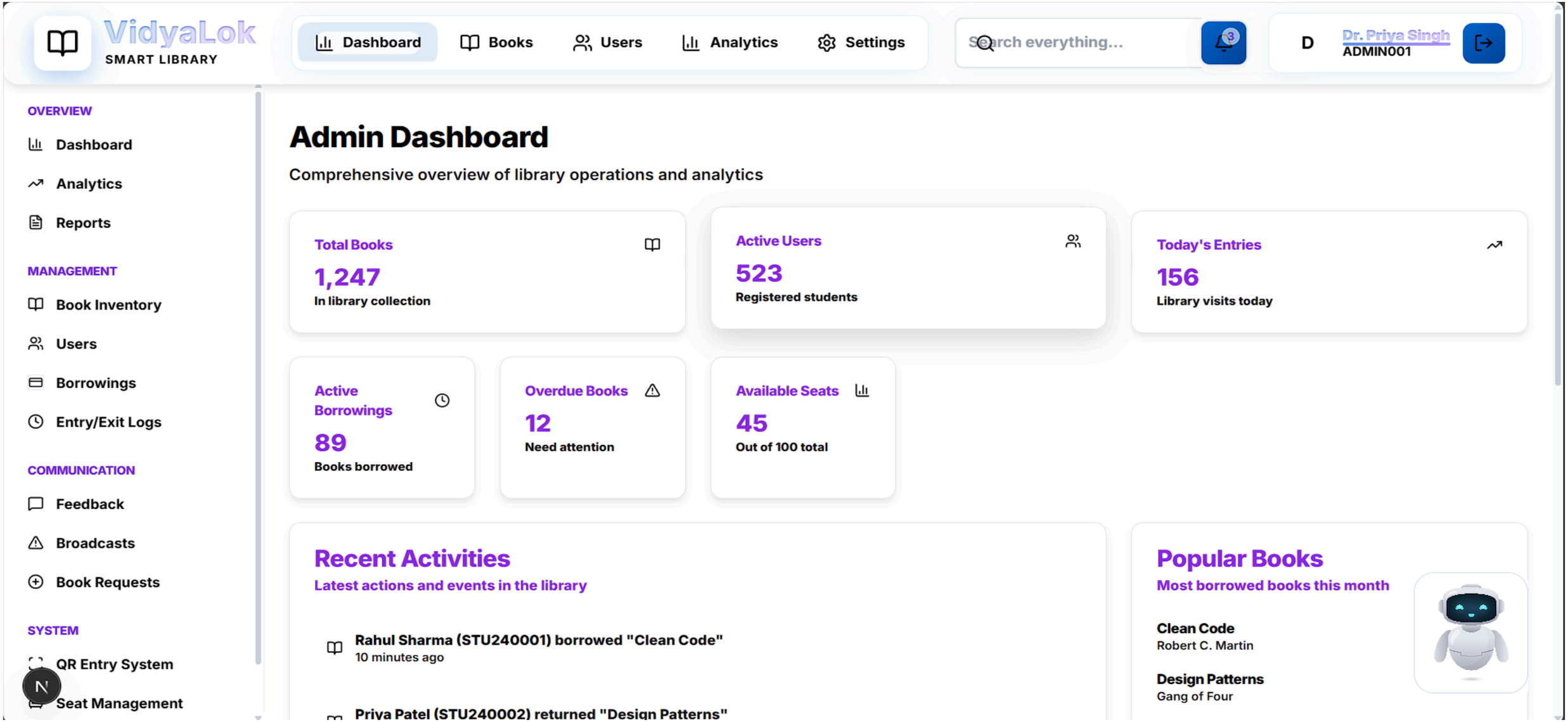

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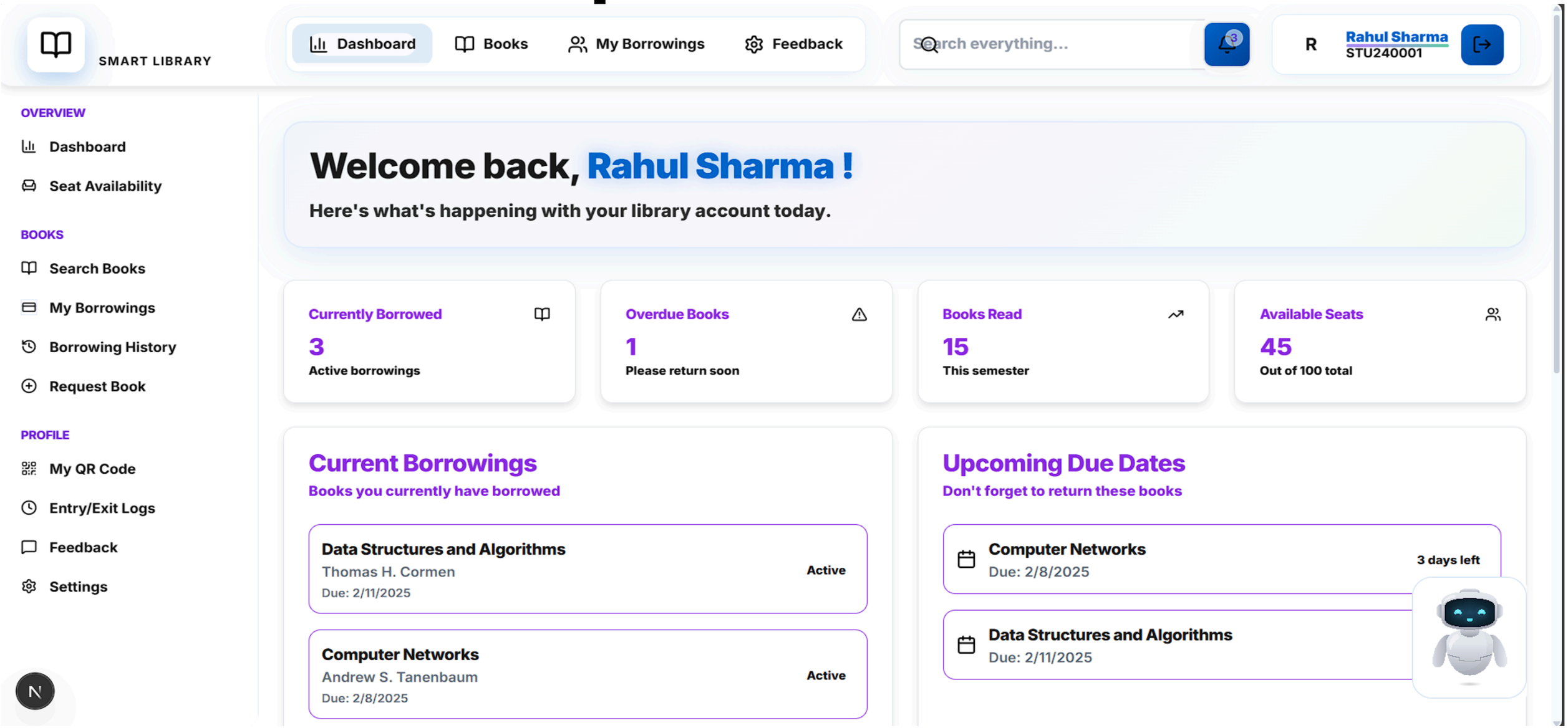

Goal Alignment

- 📘 **Quality Education (SDG 4)** – Enhances learning through intelligent book access and recommendations.

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Thank You